

APPROVED

BSc (Hons) Software Development
School of Computing, Engineering and Technology

Version	1.0
Course Type	Undergraduate
Award Type	Bachelor of Science with Honours - BSc (Hons)
Awards	
No Course Award Assigned	
Code	0738
SCQF Level	Level 10
QAA Subject Benchmark	Computing (2022)
SCQF Credit Points	480
ECTS Credit Points	240
Valid From	2027 September

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100367	computing and information technology	computing and information technology	100%
			Total 100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

	The course involves a broad range of teaching techniques, including lectures, seminars, workshops, tutorials, computer laboratories, group exercises, case studies, oral presentations, practical work, industrial placement, and a substantial project in the Honours year. The format is selected based on the suitability to the particular subject. It may consist of the staff supporting the student, engaging in problem-solving or may exist as a series of group exercises where each group is encouraged to allocate responsibilities, etc. Generally, teaching is used to discuss the underlying theories, support the peer-learning, clarify the material and experiment with the techniques and skills discussed. This may include the use of online platforms to bring interactivity and immediacy to the teaching and learning process, using platforms such as MentiMeter for quizzes, polls, word clouds, boards, or Office 365 tools for collaboratively creating boards or presentations. The ability to work as an effective member of a project team is an essential skill for anyone working in industry, and group working is a major feature of the course in Degree Stage.
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What the Course Involves	The nature of the computing elements of all courses requires the student to gain practical experience in the use of a PC or workstation. The School has multiple laboratories housing a significant number of PCs, equipped with the latest versions of open source and commercial software for the exclusive use of School of Computing, Engineering and Technology students. The computing labs are the venues where students, supported by staff, practise skills in the use of sophisticated software applications, and systems analysis and design tools in the laboratories. In each semester, students are expected to attend three to four hours per week per 15 credit module in lectures/tutorials/laboratories, with slightly more time being allocated in the early stages of the course. Delivery follows the University Academic Calendar , with timetabled sessions during the teaching weeks. Students are expected to continue their learning through independent study activities outside timetabled sessions. Each semester includes a Connect and Reflect Week, which is not used to deliver new academic content is delivered, but to provide opportunities for students to consolidate learning, reflect on progress and development goals, and have a concentrated focus on learning and development priorities. This course offers the opportunity for students to tailor their experience through complimentary themed pathways by elective choice, introduced in the common foundation year at Stage 1, and expanded upon from Stages 2 to 4. During Stages 2 and 3, the student will study two core Computing modules, and four subject modules that are core to the Software Development course. The student will elect to study a subject core module from a different course within the School, to gain an appreciation for the wider context within which Computing operates and diversify their skill set. The Placement is an important learning tool. The interaction with the staff of the company, the required tasks and the general environment provides an experience difficult to achieve by other means. Students are supported on the Placement by an academic supervisor from the staff of the School and an industrial supervisor from the company. The School supervisor supports and monitors the assessment process. In Honours Year, the student will focus on cutting-edge topics, such as integrating Artificial Intelligence into applications, developing for the latest mobile devices, and evaluating user experiences. The Capstone Project is an extended piece of work, completed under the supervision of staff from computing and other disciplines. The student agrees a specification for a task, which may be the analysis and design of a system, the development of an extensive piece of software, and produces a finished thesis.
UCAS Code	Code GW11
Course Collaborations	
No collaborations listed	
	Our BSc (Hons) Software Development course is specifically designed to create graduates who can draw on their knowledge and skills to create web and mobile-based solutions to real-world problems. This is a four year course, with the option to undertake an industrial placement or have a period studying abroad. The high level of expertise the course provides, combined with the broad range of knowledge, skills, and personal qualities which the School's graduates possess, are highly sought after. The student will develop core competencies in computing, learning to develop modern software across multiple platforms including mobile, web, desktop, and embedded systems, gaining skills in a variety of modern programming languages and frameworks. This course equips the student with the technical skills and knowledge to create modern, secure mobile and web apps that meet the challenges and requirements of the rapidly changing computing industry. Providing a core focus in object-oriented software design, the advanced development techniques that the student learns here will form the basis of their career as a professional web and app developer. In addition to providing skills that reflect the professional standards demanded by industry, the course offers the opportunity for the student to select to study from several specialised topics directly related to the active research areas of the School, including data science, artificial intelligence, cyber security, creative design, and game design.

Introduction to the Course	Stages 1 and 2 establishes essential foundations in computing, information technology, professional practice, teamwork, and individual project development. Students undertake modules shared across our computing curricula, ensuring that students build strong technical competence alongside the communication and collaborative skills expected of contemporary computing professionals. In the Degree and Honours years, the course advances this foundation through deeper technical specialisation and emerging areas of software development. Students engage with evolving approaches to intelligent and automated software construction, gaining practical experience in how AI-assisted tools are reshaping development workflows, evaluation practices, and rapid prototyping. Alongside this, they explore the design of meaningful, human-centred digital experiences, considering interaction design, accessibility, immersion, and the broader experiential dimensions of technology. In the final year, students consolidate their learning through advanced study of human–computer interaction, examining the cognitive, social, and usability factors that influence how people engage with digital systems. This emphasis ensures graduates can design, develop, and evaluate applications that are not only technically robust, but intuitive, inclusive, and responsive to real user needs. The Software Development course provides the student with both knowledge and hands-on experience in collaborating with peers across the industry. The computing industry needs confident, creative graduates with a wide, adaptable skillset, and that's what this course provides.
	Overall Aim of the Course The course level aims, which are consistent with the University Mission Statement, are to: • Produce versatile and resourceful practitioners fostering innovation, enterprise and enthusiasm for excellence in computing. • Provide the student with the knowledge and skills to equip them for a career in computing, software engineering, and information technology. • Produce practitioners/graduates who can make an effective and professional contribution in computing projects. • Develop the student's competence in requirements gathering, analysis and design, using these skills to develop computer systems that meet the clients' and end users' requirements. • Develop the student's competence in developing robust and usable software. • Assist the student in developing study and interpersonal skills required for both independent, autonomous practice and teamworking with colleagues from computing and allied subject areas. • Develop critical, analytical problem-based learning skills and the transferable skills to prepare the student for graduate employment and continuing professional development. • Provide the student with an appropriate level of knowledge, understanding and skills in the technology underpinning computer systems, computer security, communications and software development. • Provide the student with an awareness of the roles and responsibilities of a professional working within the modern computing profession, including collaborating with colleagues from other subject areas. • Provide the student with the competence to undertake computing and information technology based projects on an individual basis with minimal supervision or further training. Specifically, graduates of the BSc (Hons) Software Development course will possess the following knowledge and skills: • An understanding of the core concepts of the design and exploitation of computation and computer technology, including the importance of creative development strategies, design methodologies, artificial intelligence technologies, and innovative applications of novel technologies to software development. • The ability to apply a disciplined scientific and software engineering approach to practical problems in computing.

Educational Aims of the Course

- The ability to use and deploy current computer applications, undertake the structuring of data and information, problem identification, design development, testing and evaluation of software and computer information systems,
- The ability to design and develop robust and secure applications and systems that utilise and integrate a range of software tools, packages and platforms to provide solutions to business problems.
- A command of the human factors affecting the design of digital information systems, the judgement and decision making capabilities required to select and evaluate appropriate technologies and the skills to develop effective and usable information systems solutions to user's problems.
- The ability to critically evaluate appropriate target software platforms, including an appreciation of the uncertainty, ambiguity and trade-offs inherent in the deployment of code on these platforms, including multi-platform development approaches.
- An understanding of the relevance of theory and analysis in computing systems, including the ability to develop and use theoretical models to abstract complex processes.
- The ability to recognise the legal, social, ethical and professional issues involved in software development and be guided by the adoption of appropriate industry standards and professional practices.
- A range of professional competencies required for working in a rapidly changing computing industry, including self-management abilities (e.g. adaptability, integrity), social intelligence (e.g. collaboration, leadership), and innovation (e.g. creativity, critical thinking) developed working individually and in groups with peers from different subject areas.

The knowledge, technical, practical, and transferable skills acquired by the student during the course provides expertise in the design, implementation and evaluation of web and mobile software, across a range of platforms and computing devices. Learning equips the student with both professional skills in software development, the management of computer systems and information, and workplace collaboration, along with the academic expertise needed to pursue further study.

The skills base of this course generally resides within the body of knowledge defined by the QAA Computing benchmark statement and the ACM Computer Science curriculum. As well as the technical focus, the course all contain content that prepares the student in social, ethical, legal and professional aspects of a cooperative human environment such as the workplace.

Aims of Each Stage of the Course

Stage 1 (Certificate of Higher Education)

The student will undertake a common Foundation Year, to allow them to understand the breadth of Computing as a multidisciplinary subject, with particular focus within their chosen course. The student will acquire a knowledge and understanding of the fundamental technical concepts of programming, analysis and design of web technologies, and software development practices using appropriate tools and techniques. The student will also acquire skills, knowledge and understanding that promote continued personal and professional development through investigative, modelling, and problem solving activities that use a combination of written and computational work, both on an individual and group basis. The student will have a chance to choose to study a core module from a different course within the School to get an appreciation for the wider context within which Computing operates.

Stage 2 (Diploma of Higher Education)

The student will further develop their understanding and skills in aspects of computing that are core to software solutions. This will include advanced software design and development processes, web technologies, data structures, commonly used algorithms, and full-stack web applications. The student will also develop their understanding of research, and professional skills related to entrepreneurship and collaboration. A feature of this stage is the opportunity for students to demonstrate their ability to adapt technology to address tangible, real-world problems while considering legal, ethical, and social implications. The student will also select to study two modules from different courses within the School, thereby enabling them to establish a well-rounded skill set and knowledge base that includes topics from cybersecurity, computing and creative design, or games development, as suited for their preferred career path

Stage 3 (BSc Software Development)

The student will undertake development of computer based systems for a variety of problem domains, that require the application and synthesis of advanced programming techniques

	and design methods. This stage includes a focus on developing for a variety of platforms, learning how to use industry standard tools and frameworks in the planning, design, implementation, testing and evaluation of software solutions. The student will further develop their innovation abilities, and approaches to cooperating with colleagues from other courses in the School, while assuming an active role in the planning and implementation of software, and design projects, both as an individual and as an effective member of a small team. The student will again select to study two modules from different courses within the School, further developing their skills and knowledge based on their personal preferences and industry demand. Stage 4 (BSc (Hons) Software Development) The student will advance their knowledge of, and ability to adapt state-of-the-art technologies (including artificial intelligence, and mobile platforms) to craft solutions to complex problems. The student will learn to design and evaluate high quality user experiences for digital platforms. The student will continue to personalise their academic journey by selecting to study a module from a complementary computing course that aligns with their earlier selections. Throughout the second semester, the student will complete a capstone project, during which they will utilise an appropriate selection of tools, technologies, and techniques from at least one academic discipline to plan, design, implement and evaluate a novel solution or research artifact for a substantial and unstructured problem with due regard to user experience, human factors, client requirements, and professional standards.
Awarding Institution	Robert Gordon University
Course Duration	Blended Learning Full-Time - Full time: 4/5 years, including one year optional industrial placement in Stage 3. Exit awards at the end of each stage
	The BSc (Hons) Software Development course features an exciting and unique interdisciplinary blend of computing science, web and app development, that will enable the student to develop sophisticated applications using scientific and engineering principles. The course takes a pragmatic approach to developing the student's readiness for modern life and future employment, equipping them with the skills essential for responding to future technological and societal changes. Through the utilization of individual and group work and project teams, students are guided in cultivating: self-management abilities, such as focus, integrity, adaptability, and initiative; social intelligence by developing skills in communication, empathy, collaboration, and leadership; and innovation by fostering qualities such as curiosity, creativity, sense-making, and critical thinking.

Distinctive Features of the Course	The course has a strong practical focus on the design, implementation and testing of secure software apps and the associated information systems, along with the computer technologies that support them. A central theme of the course is multiplatform software development using multiple skills and programming languages. Students successfully completing one of the optional placement modules are awarded additional credits, and the assessments are based around a review of the student's work related skills by the industrial supervisor and academic supervisor. The course also provides the student the option to undertake a study abroad placement between Stages 2 and 3. The School partners with a number of European institutions and has recently participated in exchanges with Belgium, Finland, Norway and Sweden. This course provides students with the chance to customize their learning experience by selecting elective pathways focused on specific themes. This option is introduced in the foundation year at Stage 1 and further developed from Stages 2 to 4. In Stages 2 and 3, students will engage in two core Computing modules and four subject core modules integral to the Software Development course, opting to study a subject core module from a different course within the School. In Stage 4, the student will study three subject core modules for Software Development, one subject core module from a different computing course, culminating in the unique Capstone Project, where the student will be supported by academics from multiple disciplines, to undertake a substantial and impactful computing-focused project that harnesses the competencies throughout their degree.
Placement/Professional Experience	Students may elect to undertake a short summer placement in a summer break and/or a year-long placement between stages. The placement provides an opportunity to sample working life, be it in commerce, industry or the public sector, and apply knowledge and skills learned in the classroom to the challenges of the workplace. Students who have completed the placement as part of their course have a definite advantage when it comes to applying for graduate positions. With the combination of training, experience, and a successful degree, students are often offered subsequent employment by the placement provider with which they undertook their placement. While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who will act as the student's personal tutor during the placement. The Academic Tutor will hold a small number of scheduled video calls with the student and their Industrial Supervisor during the placement. Students may choose to undertake a short summer placement between Stages 1 and 2, or Stages 2 and 3, or Stages 3 and 4. This normally comprises at least 4 weeks of continuous, full-time supervised work experience. The assessment for a short summer placement experience is based on a review of the student's work-related skills, conducted by the Industrial Supervisor. Students on a short summer placement will normally receive at least 1 video call with their Academic Tutor. In addition, or as an alternative to a short summer placement, students may undertake a year-long placement either between Stage 2 and 3, or between Stage 3 and 4. The year-long industrial placement normally comprises of 12 months of continuous, full-time supervised work experience. The assessment for a year-long placement is based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor. Students on a year-long placement will normally receive at least two video calls with their Academic Tutor. The University provides support to students interested in undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist students by informing them of suitable placement opportunities. Students are also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or through their own connections. The Centre for Employability and Community Engagement offers further support for students, including with preparing their CV and developing interview skills.

	All placements require approval from the School prior to commencement, in order to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and students wishing to undertake a placement are expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required. The credits awarded for the placement module are in addition to the 120 credits achievable from taught modules at each stage. Successful completion is not required for progression to the following stage. The credits achieved for placement may not be used to compensate for any credits from the taught modules.
Opportunities for Further Study	An Honours degree achieved with at least an upper second class (Class 2.1) Classification will normally provide access to postgraduate and doctoral programmes in UK higher education institutions.
External and Internal Indicators of Quality and Standards	SCQF credit values have been used as a guide during the design of modules while the structure and content of the course has also been designed with the British Computer Society's (BCS) requirements for Practical/Engineering Applications in mind, and embedded in the core structure are modules and topics which provide the student with computing skills and knowledge in Underlying Theory, Design and Methodology, Tools and Techniques and Project Management and Group work. Standards are primarily established through QAA (Quality Assurance Agency) Benchmarks and the requirements imposed by the British Computer Society in order to gain professional course recognition. The achievement of appropriate standards is demonstrated through the University's course validation procedures and professional institution accreditation visits. The QAA qualification descriptors and Subject Benchmark Statements provide a reference framework that has guided the evolution of the School's programme, and the extent of the subject discipline, as defined by QAA, is broadly matched by the provision that now exists within the School's undergraduate programme. A number of internal processes and procedures are used by the School to ensure that standards are maintained and enhanced. Stakeholders' views on standards are routinely obtained from the Staff/Student Liaison Committee, Undergraduate Programme Management Team, Student Voice questionnaires, Industrial Board, industrial visits to placement students, staff involvement in professional body activities and staff participation in accreditation events outside the School and at other HE institutions. In addition to this, best practice in teaching and learning within the University is monitored through the University quality enhancement and quality assurance processes. These include the regular five-yearly cycle of Institution-Led Subject Reviews, as well as participation in the University quality assurance committee structures, such as the University's Student Experience, and Academic Regulations Sub-Committees. Finally, it should be mentioned that the lecturers within the School have an ongoing interest in pedagogical research and a commitment to expanding the scholarship of computing science education. Academic staff help to organise, and participate in, national and international conferences and have been awarded funding from the Higher Education Academy, the Scottish Informatics and Computer Science Alliance, and the Scottish Enhancement Themes to pursue work in computer science education, as well as more general work on student transitions. The School also participates in university conferences on teaching and learning.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards

University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	SQA Higher: BBCC and two further passes to include English and Maths at National 5 grade C or above if not held at Higher. GCE A Level Year 1 entry: CCC and two further passes to include English and Maths at GCSE grade 5 if not held at A Level. Irish Leaving Certificate Higher: H3H3H3H3 and two further passes to include English and Maths at grade O3 or above if not held at Higher. IB Diploma Year 1 entry: 26 to include a pass in Standard Level English and Maths at grade 4 or above if not held at Higher Level.
Alternative Qualifications for Entry	Equivalent qualifications, e.g SQA, BTEC, S/NVQs, overseas qualifications and access programmes will be considered on an individual basis. Applications from mature students with non-standard qualifications or work experience are welcomed.
Interview Procedures	Prospective applicants for the course are not normally interviewed.
Course-Specific Regulations	Honours Classification is based solely on the student's performance in Stage 4 of the course. CM4134 Capstone Project cannot be condoned [refer Academic Regulation A4, paragraph 11]. The credits achieved for placement are in addition to the 120 credits achievable from taught modules, and may not be used to compensate for any credits from the taught modules. Electives will run at the discretion of the School. Students cannot choose an elective module that they have already taken on the course, for example, some electives are aimed at particular cohorts such as articulating students. Some electives will not be available until the module is first delivered by the School, therefore, please consult the Course Leader for further information on elective availability.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU .

	All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: - To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: - To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: - To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
	Philosophy of Course Approach to Learning The BSc (Hons) Software Development course has been specifically designed to create modern software developers who can draw on their knowledge and skills to create solutions to real-world problems. Within the course there is an adherence to three principles:

- To encourage innovation through investigation and understanding of computation and computing technologies and the wider context within which these operate.
- To enhance the process of software design and development, across multiple platforms, programming languages and frameworks.
- To demonstrate the practical application of skills and knowledge in computing and engineering of applications to encourage and support creative development and problem-solving.

A feature of this course is the optional industrial or Study Abroad placement. This gives the student the opportunity to learn advanced skills while studying in a specialist university in Europe or gain invaluable work experience at an industrial partner. A distinct feature of this course is the opportunity for the student to select to study from several specialised topics directly related to the active research areas of the School, including data science, artificial intelligence, cyber security, creative design, and game design.

The course modules specifically address these ideals through the following main themes in web and mobile design.

Multi-Platform Programming

Programming is a core skill in web and mobile development. Rather than teach a single language on a single target platform the course provide the students with practical experience of software development across platforms using a number of languages. This ensures that when entering employment the graduate has had exposure to a range of development experiences and development environments. This approach encourages the students to "learn to learn" enabling them to be confident in learning new languages or technologies as they develop and as required by industry.

Security

An understanding of information security and the principles of secure development are a must for any modern software developer. A key concept of software development in the current environment of security threats is the identification of vulnerabilities in existing software and web systems. As part of the course, students will be shown the tools and techniques that are used to detect and secure vulnerable systems and websites. In addition, as software developers, the students will be required to understand the importance of including secure code in the applications they create and will be introduced to the concept of "building in" security into their own work.

Design

This theme looks at all the aspects of design that go into building software in the modern computing industry. Including aspects of software engineering, for example, the design of requirements, code implementation, and system architectures through to the design of the user-facing components of the software or system being developed. The student will be introduced to industry standard design and prototyping tools that will support and inform these processes.

Professional Skills

The final theme covers the non-technical aspects of the course. Professional skills, such as project management, communication, presentation, and writing skills are extremely desirable by the computing industry. A number of modules over the course provide the students with skills in these areas and opportunities to demonstrate them, so they are fully prepared to enter the workforce on graduation. The course culminates with the Capstone project, where each student must deliver a written thesis and present a live demonstration of their work. This allows the students' to show a full complex piece of work they have designed, documented and implemented and that they can be proud of producing.

External Competitions and Challenges

As part of the course, the students are encouraged to take part in various competitions and challenges throughout the year. These challenges include:

- The School "Hackathon" (RGUHack) and "GameJam", where students work in teams for 24-36 hours on challenges set by industry partners.
- The Google Hash Code challenge, where the students work on large optimisation challenges set by Google over a period of four hours.

Course-Specific Learning Experience

- The School of Computing has also sent teams of Computing (Application Software Development) and Computer Science students to compete in the UK Cyber Security Challenges.

The Course Team strongly believe, and student feedback has shown, that competing in these events raises students' confidence in their own ability and allows them to use the knowledge and skills they have learned in a pressure, but consequence-free, environment.

Industry Involvement

The course prides itself on its close links to industry, as part of this, some project modules feature representatives from industry partners acting as problem holders that the student design, develop, and test solutions for. We also arrange a number of opportunities for industry partners to both visit the School and host visits on their premises for our students.

Technical Facilities

Many of the practical aspects of the course take place in the computer laboratory. The School has multiple laboratories with a significant number of PCs, equipped with the latest versions of open source and commercial software for the exclusive use of School students. The School of Computing, Engineering and Technology also has a range of additional hardware for use in teaching, including iPads, Raspberry Pis, AV equipment, and VR headsets. Throughout the School, there is wireless coverage to the University network for any student wishing to use their own device.

The School's labs are regularly updated on a three year replacement cycle to ensure that the School's computing facilities are capable of delivering high performance applications with the latest software in a secured environment. Access to the labs is by electronic key fob at any time during University opening hours.

Software provision is comprehensive, including compilers for most modern programming languages and database systems. On the School's PCs, standard installed software includes a variety of development tools that changes yearly depending on taught material. Much of the software used on the course is open source or covered under an education license, allowing students to continue their work at home using the same set up as available in the labs.

	The Learning Process: Teaching Strategy and Methods The course curriculum is designed as a coherent whole with an emphasis on the integration of theory, practical skills and applications. It enables the student to acquire and develop the subject knowledge and understanding necessary for successful professional employment and supplies methodologies for the adoption of a disciplined approach to tasks such as problem definition, specification, design, implementation, deployment, and maintenance. Such knowledge includes an understanding of the range of possible options for these tasks, suitable technologies, the related practical and transferable skills including relevant approaches to collaborative work, and access to the appropriate software tools and resources. It also provides the necessary underpinning to guide practice, to ensure the sustainability of their knowledge and to provide an appropriate framework that will accommodate rapid technological change. These skills are developed more intensively at the Stage 3 (Degree Year) and Stage 4 (Honours Year) levels corresponding to the Scottish Credit and Qualifications Framework (levels 7 to 10). As the student progresses through the course they are encouraged to expand their understanding and critical appreciation of Software Development, and to extend skills in problem analysis, specification, critical design, implementation, and testing alongside maintenance into a multitude of artefacts A general feature of the course is that it stresses a unified, integrated view of the subject area. This is reflected in the fact that it is designed to:

Teaching and Learning Strategy

- Present a coherent underpinning of the theory appropriate to the aims of the course which is further developed and used throughout the period of study. This enables graduating students to adapt to future developments in the field.
- Reflect the rapid rate of change in the field and ensure that coverage is given to a selection of emerging topics so that the student are aware of likely future developments in the subject together with their potential impact.
- Integrate modes of assessment to attempt to balance the types of assessment (individual and group) and purposes of assessment (e.g. formative and summative).
- Facilitate the student to exhibit competence and achievement across a number of different themes (integrated assessment).
- Encourage the student to combine elements of their learning from different parts of the course and to show their accumulated and integrated knowledge and understanding of a topic or subject area (synoptic assessment).

The professionally orientated nature of the course requires the student to obtain general and specific skills, this being addressed by a combination of teaching and learning methods, including lectures, tutorials, computing laboratories, workshops, group exercises, case studies, oral presentations, seminars, practical work, optional placement, integrative project work, student-centred learning and private study.

Teaching and Learning Approaches

The main goal of the teaching and learning strategy is to enable students to become critical, reflective, independent learners who are active participants in the learning process. Student involvement in the learning process ensures the development of interpersonal skills and professional attitudes as well as developing the motivation and autonomy needed to take control and regulate their own educational experiences. This means that, throughout their course of study, the student will begin to take more responsibility for their own acquisition of necessary skills and knowledge so that, by the end of their course, they will be in a position to evaluate and assess their own effectiveness.

This process is driven by various objectives such as the need for constructive alignment of syllabus and learning approaches and makes use of a wide range of appropriate delivery styles. These include conventional instructional teaching methodologies, social learning activities such as group- and team-working, as well as constructivist, investigative approaches such as problem-based learning and enquiry-based learning.

As well as the day-to-day use of the latest computer hardware and software afforded by the degree, extensive use of technology-assisted learning is made within the course of study. Examples of these include assessment by means of computer-assisted assessment packages embedded in the virtual learning environment to engage the student and increase the speed of feedback and the use of an integrated approach whereby the student has short periods of instruction followed by immediate lab exercises to contextualise the topic.

Modes of Delivery

Lecture

This is usually a formal discourse for the purposes of dissemination of information, the demonstration of techniques where appropriate and the discussion of supporting ideas and consequences. The lecture is supported by a network of dissemination equipment including computer projection facilities, whiteboard, and video where appropriate. Although this type of presentation is suitable for a one-sided discourse ample opportunity exists for questions, interaction and discussion and this is facilitated by technological means.

Tutorials

This format is used for a wide range of activities, each suited to the particular subject. It may consist of the staff supporting the student engaged in problem-solving or may exist as a series of group exercises where each group is encouraged to allocate responsibilities, allocate tasks, etc. Generally, this type of teaching is used to discuss the underlying theories, support the peer-learning, clarify the material and experiment with the techniques and skills required.

Laboratory

The nature of the computing elements of the course requires the student gains practical skills in the use of a computer PC, workstation, or mobile devices. This activity takes place in one of the School's computer laboratories and consists of the student, supported by staff members, practising skills in the use of sophisticated software applications and including software development and systems analysis and design tools. There has been increasing emphasis on using a "language-lab" approach to subjects in which a series of short mini-lectures are

delivered in labs to allow the student to gain theoretical knowledge of a topic. These are interspersed with selected practical exercises, which allow this knowledge to be immediately assimilated and so reinforce the learning process.

Industrial Experience

The course includes an optional placement, which is a useful supplement to the other teaching and learning activities of the course. Students participating in the Industrial Placement have to competitively apply for placements by submitting letters of application and CVs and by attending interviews. Students on placement undertake a variety of roles and tasks for the company they are working for, supported on the Placement by a course tutor from the staff of the School and an industrial tutor from the company. The interaction with the staff of the company, the required tasks and the general environment provides an experience difficult to achieve by other means. The School tutor supports and monitors the assessment process. In addition to the Industrial Placement, there is a series of presentations and seminars given by academic staff, staff from the careers advisory service and professionals working in key positions within the Computing and Information Technology sectors of business and industry. These presentations and seminars form a key part of the CM2112 Real World Project and Professional Skills module.

Group Project

This is an extended piece of work, completed under supervision of a staff tutor. The students are assigned to small project teams and given a specification for a task, which may be the analysis, design and partial implementation of an information system, the development of a web site, mobile app or similar. The students manage themselves and work as an organised team to produce a set of deliverables, which includes the design and development of software and accompanying documentation to professional standard.

Individual Project

This is an extended piece of work, completed under supervision of one or more staff tutors. The student agrees a specification for a task, which may be the analysis and design of a system, the development of software or similar and produces a finished thesis and product. This activity replicates the development process that would occur in an industrial development or research setting.

Student's Role in Learning

The student is expected to be active participants in the learning process, becoming an active member of a learning community in which students learn from each other as well as from staff. This means that there is a requirement for students to extend their learning from formally delivered materials through wider reading, research and self-directed enquiry. As students develop over the duration of the course, they are expected to become increasingly self-directed and self-motivated, taking greater control of the study process, and defining their personal objectives for learning. Students are encouraged to reflect regularly upon their learning and to establish a view of their own progress and development, which may form the basis of tutorial guidance with staff. Students are expected to utilise the full range of facilities and resources available to them, for example the services and resources provided by the library

Assessment	The Assessment of Student Work A variety of assessment methods are used in this course, as deemed appropriate to the subject material being assessed and the context in which the material is delivered. The University has developed an Assessment and Feedback Standard, which guides all assessment practices on the course. This ensures the student has opportunities to engage with assessment and feedback which extends and enhances their knowledge and skills; offers a balanced and manageable assessment load; and provides good quality, effective feedback. Assessments are developed so that they ensure module and course-wide learning outcomes have been met corresponding at the appropriate SCQF Level 7-10 for the course stage with respect to the characteristics of knowledge and understanding; practical skills; generic cognitive skills; communication, numeracy and IT skills, autonomy, accountability and working with others. It is important to note that all grades indicated to the student during the course of the academic session are provisional until the deliberations of the Assessment Board for the course. Provisional grades are presented to this Board, which consists of academic staff involved in course delivery as well as External Examiners, and a final decision is confirmed on the performance of every student for each module undertaken by them. This is subsequently confirmed to each student in a published transcript, and may be accessed electronically on the Student Portal. The School Office prepares a schedule of coursework assessment and feedback deadlines. The assessment timetable is structured to ensure workload and deadline tensions are suitably managed. The student is informed how their assessment will be scheduled and the criteria made transparent in the context of coursework and grading structures. Work-Based Learning The optional placement is assessed by a combination of Tutor and Supervisor assessment. Computer Aided Assessment Computer-assisted assessment packages integrated into the virtual learning environment such as Moodle's question system are used in many modules, both for formative and summative assessment, to enhance student engagement and increase the speed of feedback. Peer Assessment Peer assessment is utilised in many modules throughout the course where group work is undertaken. This serves to develop working and interpersonal skills, as well as honing appraisal and critique skills. Peer assessment may be used in both summative and formative contexts. Unseen Exams In general unseen exams are used to assess the student's knowledge and understanding, and as the student progresses through the course unseen exams feature as a component of the assessment of taught modules. Assessed Laboratories Assessed laboratories are used to assess the student's knowledge, understanding and practical skills, and generally involve the student undertaking an unseen practical task such as developing a part of an information system, under open or closed book, supervised conditions with a fixed time constraint. Coursework For summative assessment the achievement of Learning Outcomes is assessed by coursework or a combination of exam and coursework. Coursework may take many forms, i.e. reports, essays, production of software, portfolio of work, presentations, design documentation and modelling activities. Projects/Dissertations Extended projects are used in later stages as a learning vehicle and assessment process that allow the student to tackle more complex problems that often require the integration of knowledge and understanding, and intellectual skills from a variety of subject areas. These projects often require the practical application of a number of different techniques and technologies and provide the students with the opportunity to learn new investigative and evaluative skills. Projects, in particular group projects, provide an opportunity to learn new investigative and evaluative skills, and are also often used as the vehicle for gaining valuable employability skills. Project work is usually assessed by a combination of assessment of the completed project deliverable as an artefact, accompanying report or dissertation, and verbal presentation and demonstration documenting the project work undertaken by the student.

	Assessment Types Some modules use a combination of assessment types that integrate modes of assessment in order to balance the types of activity (e.g. individual and group activities) and purposes of assessment (e.g. formative and summative). Where there are a number of modules that cover allied subject material, integrated assessment may be used to facilitate the student to exhibit competence and achievement across a number of different themes. In all years, synoptic assessments embedded into projects and other practical assignments are used to encourage the student to combine elements of their learning from different parts of the course and to show their accumulated knowledge and understanding of a set of topics or a subject area. A student's Honours Classification (First Class Honours, Upper Second Class Honours etc) is based solely on performance in Stage 4 of the course. Equality and Diversity Any datasets used within the course will aim to adhere to the Software Engineering Code of Ethics and Professional Practice recommended by the ACM/IEEE-CS Joint Task Force which states that "only accurate data derived by ethical and lawful means" will be used, and that its integrity will be maintained. Furthermore, the student will be taught about recognising and researching known bias within datasets, and where possible, will be signposted to use datasets that are less likely to have inherent bias.
University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • an ongoing scheme of personal/pastoral support for students, including online support, provided by the <i>Inclusion Centre</i>; • an extensive programme of student study skills delivered through <i>Student Life</i> and the <i>Library</i>; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in the Quality Assurance Agency's Quality Enhancement Themes, enhancementthemes.ac.uk ; • support for staff from the <i>Department of Academic Quality and Learning Enhancement (AQLE)</i>, including credit rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk/ , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
	In addition to the University-wide support for Teaching and Learning, the school provides a number of means of supporting the student in teaching and learning. Members of staff publish course based teaching and learning materials on the University's Virtual Learning Environment, Moodle. The range of published materials spans from supplementary teaching materials such as additional lecture notes and tutorial guides to software development to general guidance for the student, such as codes of conduct in laboratories. Information on studying abroad through the Erasmus scheme can be accessed at http://www.rgu.ac.uk/future-students/study-abroad .

Course-Specific Support for Teaching, Learning and Assessment	The school runs a comprehensive induction programme during freshers week where the student will be introduced to the school, introduced to key members of staff involved in their course. During the induction event, the student will be made familiar with the University's systems. The students can also request keyfob access to dedicated labs and other spaces where teaching takes place. Feedback is provided continuously to the student through informal contact with subject lecturers and tutors in the tutorial and laboratory settings. Feedback is provided on assessed practical work within the universities standard feedback window. Feedback on examination results can be provided by module staff, and facilities are made available for all students to view their marked examination scripts on request and under supervision. Grades are made available electronically for viewing via the student portal following ratification by the Assessment Board. Details on the School, course, and working environment are available through our VLE, Moodle. It contains information regarding teaching and assessment, IT support, student support services, learning resources including the library and health & safety. It also details key contact names and email addresses where appropriate.
University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	This course fully complies with the above procedures. The course management team meets regularly to review any specific course issues, student feedback and module evaluation, External Examiners' reports and matters arising from Staff Student Liaison meetings. These meetings will include representatives from the School of Computing, Engineering and Technology. Annual Course/Programme Appraisals are reviewed and approved by the Programme Management Team and School Academic Board which meet at least once per semester. Within the School, the School's Teaching Committee regularly looks at module metrics such as student experience and student achievement to ensure that the models are continuing to deliver the required standard set by the school. The School's Teaching Committee, the Programme Management Team and School Academic Board also review all proposed changes to the structure and content of the course. The Undergraduate Computing Programme is also informed by feedback from the School's Industrial Board.

University Student Feedback Mechanisms	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements. This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnairesand where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level; • support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
Course-Specific Student Feedback Mechanisms	Within the School, the Staff/Student Liaison Committee for all undergraduate degree courses normally consists of representatives from the School management and Course Leaders, with student representatives from each stage of each course. The Committee meets at least once per semester and feeds appropriate information to other departments and/or committees as required. Students can take a role as a School Officer allowing them participation in the School Academic Board to contribute to discussions about School policy. In addition, there are regular scheduled meetings between the student and Course Leaders to discuss issues arising within specific classes and on individual modules. Student Voice questionnaires are distributed to students on all modules during semester which allow them to raise issues and give feedback to staff.

Awards

Stage 1

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Certificate of Higher Education - CertHE	Computing	Aberdeen, United Kingdom	Exit Award		120

Stage 2

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Diploma of Higher Education - DipHE	Computing	Aberdeen, United Kingdom	Exit Award		240

Stage 3

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Bachelor of Science - BSc	Software Development	Aberdeen, United Kingdom	Exit Award		360

Stage 4

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Bachelor of Science with Honours - BSc (Hons)	Software Development	Aberdeen, United Kingdom	Target Award		480

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Describe basic knowledge and understanding of the basic context of computer science, and its relationship to other subject areas such as mathematics, science and engineering.	SC1	SCQF 7
LLO2	Outline familiarity with the basic facts and principles of computing and supporting topics via demonstrations of skills gained, the ability to answer questions and to describe and employ basic tools and techniques. The student is introduced to the basic skills and knowledge required for operating in a computing environment via study and practice. The student should understand, at an appropriate, elementary level, programming and the development of software, computer hardware, and database systems, and show an understanding of the business environment and roles played by information systems in organisations.	SC1	SCQF 7
LLO3	Carry out study of the subject in a systematic, directed way with the aid of appropriate tutor guidance within classroom and laboratory.	SC2	SCQF 7
LLO4	Apply the knowledge, skills and understanding of problem solving, software development, design, and computer science in practical contexts.	SC2	SCQF 7
LLO5	Employ some of the basic and routine professional skills, techniques, and practices associated with modern computing.	SC2	SCQF 7
LLO6	Apply these skills in both routine and non-routine contexts.	SC2	SCQF 7
LLO7	Apply the basic principles and procedures found in computing science to standard, simple situations.	SC3	SCQF 7
LLO8	Present the ability to analyse situations and problems, based on the set of classifications presented by tutors.	SC3	SCQF 7
LLO9	Systematically relate to a limited number of facts/ideas/elements in an imitative manner.	SC3	SCQF 7
LLO10	Start to define their own value judgments based on the criteria provided.	SC3	SCQF 7
LLO11	Apply the basic tools/methods of computing to simple, standard problems.	SC3	SCQF 7
LLO12	Convey aspects of computer science in a clear and concise way, in writing and orally, in relatively informal and limited-length pieces of work. The student should also be able to communicate appropriately the basic design and documentation of programs.	SC4	SCQF 7
LLO13	Achieve basic numeracy, mathematics and the ability to manipulate data at a level appropriate to this stage of the degree. The student should also develop an understanding of basic concepts in logic and computation.	SC4	SCQF 7
LLO14	Obtain practical skills in computer science using basic ICT applications as well as those skills necessary for efficient use of computer networks.	SC4	SCQF 7
LLO15	Manage learning in an increasingly effective and purposeful way, showing the beginnings of development as an autonomous learner, and exercise some initiative and independence in carrying out defined activities at a professional level of practice.	SC5	SCQF 7
LLO16	Carry out interactions effectively with tutors and fellow students via software and web development tasks, problem-solving tutorials and the production of appropriate documentation, and accept supervision in less familiar areas of work.	SC5	SCQF 7
LLO17	Exercise some managerial or supervisory responsibility for the work of others within a defined and supervised structure while managing limited resources within defined areas of work.	SC5	SCQF 7
LLO18	Take the lead in implementing agreed plans in familiar or defined contexts, while taking account of own and others' roles and responsibilities when carrying out and evaluating tasks.	SC5	SCQF 7
LLO19	Work, under guidance, with others to acquire an understanding of current professional practice	SC5	SCQF 7
LLO20	Report an understanding of the important central areas of this stage of the course, specifically, software development, dynamic web development, data structures and algorithms, collaborative and professional skills, and research methods.	SC1	SCQF 8

LLO21	Infer a coherent picture of the discipline by furthering their appreciation of the scope of the field of study.	SC1	SCQF 8
LLO22	Compare the significance of central and peripheral areas of subject, suggesting opportunities to explore and extend the boundaries of the discipline.	SC1	SCQF 8
LLO23	Practice knowledge, skills and understanding in using a range of professional skills, techniques, and practices associated with the topics studied, some of which are at an advanced or complex level.	SC2	SCQF 8
LLO24	Show the ability to carry out routine lines of enquiry, development or investigation into professional level problems and issues in relation to web and mobile systems.	SC2	SCQF 8
LLO25	Adapt routine practices within accepted standards.	SC2	SCQF 8
LLO26	Use standard principles and procedures to somewhat more demanding situations.	SC3	SCQF 8
LLO27	Demonstrate increasing ability to undertake classifications and critically analyse relatively simple situations.	SC3	SCQF 8
LLO28	Demonstrate an appreciation of the need for relating and marshalling a range of facts/ideas/elements in an argued case.	SC3	SCQF 8
LLO29	Compare a range of valid alternative solutions, and begin to discriminate and evaluate in a reasoned, systematic and increasingly independent way, particularly within the context of group work.	SC3	SCQF 8
LLO30	Use given tools/methods accurately and carefully to more demanding problems appropriate to this stage.	SC3	SCQF 8
LLO31	Use communication skills in more formal academic and professional styles, and in more extended pieces of work. Some presentation work may be introduced, including through the optional placement process.	SC4	SCQF 8
LLO32	Undertake the application of a selection of more advanced numerical, modelling and design skills, relevant to specialised areas of software development, dynamic web programming, and data structures and algorithms.	SC4	SCQF 8
LLO33	Use a broad selection of more advanced computing and IT skills in a wide range of contexts in all topics studied.	SC4	SCQF 8
LLO34	Practice effective interactions within a learning- or subject-specific group, demonstrating basic negotiating, role-ownership, leadership and group-support skills.	SC5	SCQF 8
LLO35	Plan work activities in familiar or defined contexts.	SC5	SCQF 8
LLO36	Practice management of resources and exercise responsibility for the work of others within a group, within a defined structure. Practice in ways that show awareness of own and others' roles, responsibilities and contributions when carrying out and evaluating tasks.	SC5	SCQF 8
LLO37	Work, under guidance, with others to acquire an understanding of current legal, ethical and professional issues, and manage these issues in accordance with current professional and/or ethical codes of practices.	SC5	SCQF 8
LLO38	Discuss important central areas of this stage of the degree; project management, product creation, mobile development, marketing, and visual communication.	SC1	SCQF 9
LLO39	Explain a range of the principles, main theories, concepts and terminology in the web and mobile area, including, communication, development, security, and marketing.	SC1	SCQF 9
LLO40	Discuss the broad areas of the knowledge/subject bases relevant to modern web and mobile systems, including key specialist areas.	SC1	SCQF 9
LLO41	Explain the limits of the broad area of study, together with an ability to relate the different areas to one another in a systematic and logical manner.	SC1	SCQF 9
LLO42	Demonstrate knowledge, skills and understanding in using a range of the principal professional skills, techniques, practices and/or materials associated with web and mobile systems.	SC2	SCQF 9

LLO43	Demonstrate an applied knowledge of some specialised or advanced skills, techniques, practices and/or materials in the design, implementation, communication, and marketing of web and mobile solutions.	SC2	SCQF 9
LLO44	Demonstrate application of routine methods of enquiry and research.	SC2	SCQF 9
LLO45	Demonstrate ability to work in a range of professional contexts that include a degree of unpredictability.	SC2	SCQF 9
LLO46	Demonstrate ability to select and use principles and procedures appropriate to the situation or problem in hand.	SC3	SCQF 9
LLO47	Point out familiar ideas or principles in new contexts or situations; analyse systematically and effectively.	SC3	SCQF 9
LLO48	Assemble facts/ideas/elements in an argued case or design in a wide range of situations, particularly within software project management.	SC3	SCQF 9
LLO49	Differentiate among a range of valid alternative solutions; begin to discriminate and evaluate in a reasoned, systematic and increasingly independent way using a range of sources.	SC3	SCQF 9
LLO50	Draw on established problem-solving skills to begin self-direct problem solving, including differentiating key elements or route professional problems, and organising appropriate methods to construct their resolution, with only general guidance in all topics.	SC3	SCQF 9
LLO51	Effectively interpret communication requirements for variety of roles, to produce clear, well-structured presentations and reports utilising appropriate design principles, on relevant aspects of web and mobile systems.	SC4	SCQF 9
LLO52	Defend the selection of specialist numerical, modelling and design skills used in the development of secure web and mobile systems, and the management of web and mobile development projects.	SC4	SCQF 9
LLO53	Access and successfully use a broad selection of specialist computing and ICT skills in a wide range of contexts, in all topics.	SC4	SCQF 9
LLO54	Demonstrate a broad-ranging and flexible approach to study, identifying learning needs and pursuing activities designed to meet these needs in increasingly autonomous ways.	SC5	SCQF 9
LLO55	Assume an appropriate role in the design and management of web and mobile development projects.	SC5	SCQF 9
LLO56	Demonstrate ability to use negotiation and collaboration skills to manage the interface between computing, design, marketing, and other functional areas of an organisation, and the users of its computer systems.	SC5	SCQF 9
LLO57	Assess appropriate response to legal, ethical and professional issues in accordance with current codes of best practice, seeking guidance where appropriate.	SC5	SCQF 9
LLO58	Illustrate an understanding of the context of, and relationship between, the principal areas, features, boundaries, terminology and conventions of specialist areas of the course. The specialist areas at this level are: Advanced Mobile Development, Artificial Intelligence, and User Experience.	SC1	SCQF 10
LLO59	Illustrate confident familiarity and critical understanding of the defining concepts and features of web and mobile information systems.	SC1	SCQF 10
LLO60	Develop appropriate high-level expertise in relevant implementation skills, user experience, and artificial intelligence techniques, and apply specialist analysis within the Capstone Project, while learning technical and design skills appropriate to the specific task.	SC1	SCQF 10
LLO61	Communicate detailed knowledge and understanding in one or more specialisms related to web and mobile systems, some of which are informed by, or at the forefront of the subject.	SC1	SCQF 10
LLO62	Conceptualise a clear understanding of their subject's boundaries, of the limits of its application, and of the range of methods of study and types of judgment employed.	SC1	SCQF 10
LLO63	Execute knowledge, skills and understanding in using a wide range of the principal professional skills, techniques, and practices associated with web and mobile design and development.	SC2	SCQF 10
	Illustrate practical skills in the use of techniques and materials that are specialised, advanced and at the forefront of web and mobile information		

LLO64	systems.	SC2	SCQF 10
LLO65	Illustrate application of practical knowledge relevant to web and mobile information systems to execute a defined project of research, design, development or investigation in the Capstone Project module.	SC2	SCQF 10
LLO66	Operate in a range of professional level contexts that include a greater degree of unpredictability and specialism in web and mobile information systems.	SC2	SCQF 10
LLO67	Compose an accurate selection and application of principles and procedures to the solution of a range of professional situations and problems, working autonomously, particularly within the Capstone Project module.	SC3	SCQF 10
LLO68	Examine principles and ideas in new contexts and situations, analysing them systematically, effectively and critically, particularly within the Capstone Project module.	SC3	SCQF 10
LLO69	Compose facts, ideas and elements in support of a well-structured argument, design novel solutions and evolve new concepts, working autonomously, particularly within the Capstone Project module.	SC3	SCQF 10
LLO70	Reconcile theory with professional and vocational practice, autonomously evaluate theory, process, solutions and outcomes critically and effectively, particularly within the Capstone Project.	SC3	SCQF 10
LLO71	Develop creative solutions to complex problems in an increasingly independent, confident and flexible way and apply knowledge and skills appropriate to their solution, at the threshold of professional competence in all topics, but, particularly within the Capstone Project module.	SC3	SCQF 10
LLO72	Justify the use of a broad selection of specialist software engineering, artificial intelligence, communication, marketing, and information systems skills, in all topics, but particularly within the Capstone Project module.	SC4	SCQF 10
LLO73	Communicate their effective use of specialist computing, design, and ICT facilities concerned with mobile and web software development and information systems and develop appropriate skills for this purpose.	SC4	SCQF 10
LLO74	Engage effectively in independent roles, argue in a professional manner, and produce detailed critiques and coherent project reports.	SC4	SCQF 10
LLO75	Communicate confidently with informed audiences in a range of contexts via oral and formal presentations, particularly within the Capstone Project.	SC4	SCQF 10
LLO76	Critique the demonstration of their skills, knowledge and understanding through autonomous investigative, modelling and problem-solving activities, and management of the process relating to the development, maintenance and review of mobile, and web information systems.	SC5	SCQF 10
LLO77	Argue the extent to which they maintain standards and manage complex ethical and professional issues and make informed judgments on issues not addressed by current professional and/or ethical codes or practices.	SC5	SCQF 10
LLO78	Critique their own approach to independent learning when provided with minimal guidance and using a wide range of resources appropriate to the profession, seeking and making effective use of feedback, leading to personal development. At this level within all topics, but particularly within the Capstone Project.	SC5	SCQF 10
LLO79	Interact effectively within groups in all topics, and, where appropriate, demonstrate leadership and/or initiative.	SC5	SCQF 10

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10

		SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Stage 1 / Semester 1**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM1109	Exploring Computing Devices		Tiffany Young (ty7997)	Yes	2.0	15	Level 7
Full Time On Campus	CM1119	Problem Solving and Maths		Mark Snaith (ms10541)	Yes	2.0	15	Level 7
Full Time On Campus	CE1337	Programming Bootcamp		Mark Snaith (ms10541)	Yes	1.0	15	Level 7
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0738 - Electives 1	Software Development - Electives Group 1		Laura Tucker		1.0	15	

Stage 1 / Semester 2**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>

Full Time On Campus	CM1120	Database Systems		Tiffany Young (ty7997)	Yes	3.0	15	Level 7
Full Time On Campus	CM1113	Software Design and Development		Mark Snaith (ms10541)	Yes	3.0	15	Level 7
Full Time On Campus	CM1121	Web UX		Jay Munro	Yes	3.0	15	Level 7
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0738 - Electives 2	Software Development - Electives Group 2		Laura Tucker		1.0	15	

Stage 2 / Semester 1**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM2115	Advanced Software Design and Development		Mark Snaith (ms10541)	Yes	1.0	15	Level 8
Full Time On Campus	CE2000	Dynamic Web Technologies		David Corsar (dc9396)	Yes	1.0	15	Level 8
Full Time On Campus	CM2112	Real World Projects and Professional Skills		Ross McLean (rjm7342)	Yes	1.0	15	Level 8
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0738 - Electives 3	Software Development - Electives Group 3		Laura Tucker		1.0	15	

Stage 2 / Semester 2**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM2116	Data Structures and Algorithms		David Corsar (dc9396)	Yes	2.0	15	Level 8
Full								

Time On Campus	CE2001	Full Stack Applications		David Corsar (dc9396)	Yes	1.0	15	Level 8
Full Time On Campus	CM2124	Research Methods		Daniel Gardner	Yes	2.0	15	Level 8
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0738 - Electives 4	Software Development - Electives Group 4		Laura Tucker		1.0	15	

Stage 2 / Semester 3**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0738 - Electives 5	Software Development - Electives Group 5		Ross McLean (rjm7342)		1.0	90	

Stage 3 / Semester 1**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE3026	Autonomous Coding		Ross McLean (rjm7342)	Yes	1.0	15	Level 9
Full Time On Campus	CM3130	Interdisciplinary Team Project		Mark Zarb (mz7371)	Yes	1.0	15	Level 9
Full Time On Campus	CM3147	Native Mobile Apps		David Corsar (dc9396)	Yes	1.0	15	Level 9

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0738 - Electives 6	Software Development - Electives Group 6		Laura Tucker		1.0	15	

Stage 3 / Semester 2**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE3025	Experiential Development		Jay Munro	Yes	1.0	15	Level 9
Full Time On Campus	CM3138	Hybrid Apps		David Corsar (dc9396)	Yes	1.0	15	Level 9
Full Time On Campus	CM3141	Pirate Studies		Ross McLean (rjm7342)	Yes	1.0	15	Level 9
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0738 - Electives 7	Software Development - Electives Group 7		Laura Tucker		1.0	15	

Stage 3 / Semester 3**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0738 - Electives 5	Software Development - Electives Group 5		Ross McLean (rjm7342)		1.0	90	

Stage 4 / Semester 1**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM4133	Cloud AI		David Corsar (dc9396)	Yes	1.0	15	Level 10
Full Time On Campus	CB3162	Digital Media And User Experience		Josh Bircham (jb10079)	Yes	1.0	15	Level 9
Full Time On Campus	CM4110	Human Computer Interaction			Yes	1.0	15	Level 10
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0738 - Electives 8	Software Development - Electives Group 8		Laura Tucker		1.0	15	

Stage 4 / Semester 2**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE4321	Capstone Project		Mark Zarb (mz7371)	Yes	1.0	60	Level 10

Course Deliveries

Stage	Semester	Delivery Type	Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
1	Semester 1	Core	Full Time On Campus	CM1109	Exploring Computing Devices		Tiffany Young (ty7997)	Yes	2.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CM1119	Problem Solving and Maths		Mark Snaith (ms10541)	Yes	2.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CE1337	Programming Bootcamp		Mark Snaith (ms10541)	Yes	1.0	15	Level 7
1	Semester 1	Elective	Full Time On Campus	0738 - Electives 1	Software Development - Electives Group 1		Laura Tucker		1.0	15	
1	Semester 2	Core	Full Time On Campus	CM1120	Database Systems		Tiffany Young (ty7997)	Yes	3.0	15	Level 7
1	Semester 2	Core	Full Time On Campus	CM1113	Software Design and Development		Mark Snaith (ms10541)	Yes	3.0	15	Level 7
1	Semester 2	Core	Full Time On Campus	CM1121	Web UX		Jay Munro	Yes	3.0	15	Level 7
1	Semester 2	Elective	Full Time On Campus	0738 - Electives 2	Software Development - Electives Group 2		Laura Tucker		1.0	15	
2	Semester 1	Core	Full Time On	CM2115	Advanced Software Design and Development		Mark Snaith (ms10541)	Yes	1.0	15	Level 8

			Campus								
2	Semester 1	Core	Full Time On Campus	CE2000	Dynamic Web Technologies		David Corsar (dc9396)	Yes	1.0	15	Level 8
2	Semester 1	Core	Full Time On Campus	CM2112	Real World Projects and Professional Skills		Ross McLean (rjm7342)	Yes	1.0	15	Level 8
2	Semester 1	Elective	Full Time On Campus	0738 - Electives 3	Software Development - Electives Group 3		Laura Tucker		1.0	15	
2	Semester 2	Core	Full Time On Campus	CM2116	Data Structures and Algorithms		David Corsar (dc9396)	Yes	2.0	15	Level 8
2	Semester 2	Core	Full Time On Campus	CE2001	Full Stack Applications		David Corsar (dc9396)	Yes	1.0	15	Level 8
2	Semester 2	Core	Full Time On Campus	CM2124	Research Methods		Daniel Gardner	Yes	2.0	15	Level 8
2	Semester 2	Elective	Full Time On Campus	0738 - Electives 4	Software Development - Electives Group 4		Laura Tucker		1.0	15	
2	Semester 3	Elective	Full Time On Campus	0738 - Electives 5	Software Development - Electives Group 5		Ross McLean (rjm7342)		1.0	90	
3	Semester 1	Core	Full Time On Campus	CE3026	Autonomous Coding		Ross McLean (rjm7342)	Yes	1.0	15	Level 9
			Full								

3	Semester 1	Core	Time On Campus	CM3130	Interdisciplinary Team Project		Mark Zarb (mz7371)	Yes	1.0	15	Level 9
3	Semester 1	Core	Full Time On Campus	CM3147	Native Mobile Apps		David Corsar (dc9396)	Yes	1.0	15	Level 9
3	Semester 1	Elective	Full Time On Campus	0738 - Electives 6	Software Development - Electives Group 6		Laura Tucker		1.0	15	
3	Semester 2	Core	Full Time On Campus	CE3025	Experiential Development		Jay Munro	Yes	1.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CM3138	Hybrid Apps		David Corsar (dc9396)	Yes	1.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CM3141	Pirate Studies		Ross McLean (rjm7342)	Yes	1.0	15	Level 9
3	Semester 2	Elective	Full Time On Campus	0738 - Electives 7	Software Development - Electives Group 7		Laura Tucker		1.0	15	
3	Semester 3	Elective	Full Time On Campus	0738 - Electives 5	Software Development - Electives Group 5		Ross McLean (rjm7342)		1.0	90	
4	Semester 1	Core	Full Time On Campus	CM4133	Cloud AI		David Corsar (dc9396)	Yes	1.0	15	Level 10
4	Semester 1	Core	Full Time On Campus	CB3162	Digital Media And User Experience		Josh Bircham (jb10079)	Yes	1.0	15	Level 9

			Campus								
4	Semester 1	Core	Full Time On Campus	CM4110	Human Computer Interaction			Yes	1.0	15	Level 10
4	Semester 1	Elective	Full Time On Campus	0738 - Electives 8	Software Development - Electives Group 8		Laura Tucker		1.0	15	
4	Semester 2	Core	Full Time On Campus	CE4321	Capstone Project		Mark Zarb (mz7371)	Yes	1.0	60	Level 10

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	On-Campus	Full Time	

Sources of Further Information

#	URL and Description
1	https://www.bcs.org/media/9006/course-accreditation-guidelines.pdf A number of external sources were consulted in the process of curriculum development for the BSc (Hons) Web and Mobile Design course. The accreditation guidelines for the BCS, The Chartered Institute for IT, Guidelines on course accreditation, Information for universities and colleges, January 2020, can be found at https://www.bcs.org/media/9006/course-accreditation-guidelines.pdf and Academic Accreditation Guidelines, April 2022 can be found at https://www.bcs.org/media/1209/accreditation-guidelines.pdf The Association for Computing Machinery (ACM) Guidelines on curricular elements for Computer Science (and related areas), Computer Science Curricula 2020, can be found at https://www.acm.org/binaries/content/assets/education/curricula-recommendations/cc2020.pdf